

Experimental Evaluation of 5G NR OFDM-Based Passive Radar Exploiting Reference, Control, and User Data

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Abstract

In communication-centric Integrated Sensing and Communication (ISAC) systems, passive radars utilize existing communication signals of opportunity for sensing. The conventional approach relies on computing the cross-ambiguity function (CAF) between reference and surveillance signals to generate delay-Doppler/range-velocity maps (DDM/RVM). In contrast, modern OFDM-based sensing approach exploits the Channel Frequency Response (CFR), originally estimated in the communication receiver for equalization, to produce radar RVMs. In OFDM-based passive radars utilizing 4G LTE or 5G NR waveforms, CFR estimation typically uses only reference signals. However, simulation-based studies, often assuming a priori knowledge of user data symbols, suggest potential performance gains from also incorporating user data and control channels. In this work, we present experimental results from an ISAC receiver that jointly utilizes all components of the 5G NR downlink waveform, synchronization signals (PSS, SSS), control channels (PBCH, PDCCH), data channels (PDSCH), and reference signals (PBCH DM-RS, PDCCH DM-RS, PDSCH DM-RS, CSI-RS), for CFR estimation and RVM generation. We further perform communication processing and evaluate the Bit Error Rate (BER) of user data transmission. Detection performance for vehicles moving on a high-speed road is compared across CSI-RS-only, extended control/data utilization, and classical CAF-based radars. Results demonstrate that combining user data and control information significantly enhances sensing accuracy and detection robustness over pilot-only approaches.

Keywords: 5G New Radio, illumination of opportunity, passive radar, delay-Doppler map (DDM), range-velocity map (RVM), channel frequency response (CFR), channel impulse response (CIR), reference data, control data, user data

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1. Introduction

The concept of utilizing a single waveform for both radar and communication functions has been extensively studied for many years, with numerous key milestones and a rich research history [1,2]. Three primary designs have been identified in this domain: radar-centric, communication-centric, and joint designs [3,4]. Given the broad scope of research in this area, our focus is limited to the communication-centric approach. In the communications, Integrated Sensing and Communication (ISAC) has attracted significant attention, as it is envisioned to be one of the key enablers of 6G networks [3–5]. Moreover, it has already been incorporated as a study item [6] within the 3GPP standardization efforts for 5G, underscoring its growing importance in future wireless systems.

Numerous studies have explored various approaches to utilizing 5G signals as illuminators of opportunity for radar systems [CYTOWANIA]. Reference [7] discusses several applications, including active sensing, where the base station both transmits and receives radar pulses, and passive sensing, where an independent ISAC receiver listens to the 5G downlink and uses it for target detection. In passive detection [8], the predominant technique is based on the cross-ambiguity function (CAF), which relies on correlating the received 5G downlink signal, reflected from potential targets, with the line-of-sight (LOS) signal recorded by a separate antenna (e.g., [9]).

An alternative approach uses the fact that in modern wireless communication systems—such as Digital Audio Broadcasting (DAB), Digital Video Broadcasting-Terrestrial (DVB-T/T2), Wi-Fi, and 4G/5G, the orthogonal frequency division multiplexing (OFDM) [10] is the dominant transmission technology. In OFDM systems, in addition to user and control channels, the base station transmits reference signals, sequences known to the receiver. These are used to estimate the channel frequency response (CFR) and to compensate for channel effects during data demodulation. This waveform design enables an OFDM-based sensing technique [11,12], in which the CFR is further analyzed to extract information about moving objects in the vicinity of the receiver.

Several studies have demonstrated sensing results for OFDM-based radars utilizing 5G downlink reference signals. In [13], an overview of various 5G downlink pilots and their applicability to sensing is presented. [14,15] illustrate how 5G Positioning Reference Signals (PRS) can be effectively utilized for radar purposes. Furthermore, in our previous works [16,17], we demonstrated that 5G signals including Channel State Information Reference Signals (CSI-RS) and custom Radar-on-Demand (RoD) pilots can be used to detect moving vehicles, both in simulations and real-world field measurements.

Although reference signals are promising candidates for introducing sensing capabilities into existing communication infrastructures, they are typically sparsely distributed within the OFDM grid. This sparsity introduces several limitations, such as a reduced maximum measurable velocity and lower signal energy available for target illumination. Since communication networks are primarily designed to transmit user data, these data payloads occupy most of the OFDM resources. This poses a significant challenge, as unlike pilots, which are pseudo-random sequences with well-defined correlation properties, user data consists of random sequences that are unknown in bistatic configurations. Additionally, repurposing data payloads and control channels for sensing can result in non-uniform sampling of the CFR.

To address these challenges, several studies have been conducted. [18] provides a technical overview of sensing and communication (S & C) using both reference signals and user data. In [19,20], sensing in a monostatic configuration or bistatic configuration with perfect decoding is evaluated with a priori knowledge of transmitted user bits. References [21,22] analyze the sensing characteristics of various 5G synchronization and pilot signals, as well as data payloads, and presents simulation results for vehicle detection. In [23], an iterative OFDM-based sensing approach for multi-target scenarios that uses both pilots and user data is proposed. Furthermore, in our previous work [24], we demonstrated that repurposing user data OFDM resources for radar applications can enhance sensing performance. Through simulations, we investigated the effects of different QAM modulation orders of data payloads and examined how transmission errors, quantified by bit error rate (BER), influence radar sensing capabilities.

The aim of this paper is to further investigate the potential of exploiting all components of the 5G downlink waveform for sensing purposes, including primary and secondary synchronization signals (PSS and SSS), physical broadcast channel (PBCH), physical downlink control channel (PDCCH), channel state information reference signals (CSI-RS), demodula-

tion reference signals (DM-RS), and user data, i.e. the physical downlink shared channel (PDSCH). Unlike most existing studies that analyze this concept primarily through simulations, we conducted real-world field measurements in which three different 5G signals were transmitted and recorded by a receiver positioned near a high-speed road. Then, we performed radar processing to compare the performance of OFDM-based radars utilizing only CSI-RS reference signals with those using user data and/or additional control channels, as well as with the conventional CAF-based radar. In contrast to many prior works, we did not assume a priori knowledge of user data payloads. Instead, we carried out communication demodulation and decoding of the transmitted signals, allowing us to measure BER under which user data symbols were reconstructed.

The remainder of this paper is organized as follows. Section 2 presents the mathematical foundation of OFDM-based sensing in a fading channel. Section 3 describes the 5G ISAC receiver architecture, including the communication subsystem (Subsection 3.1), the OFDM-based sensing module (Subsection 3.2), and the CAF-based sensing module (Subsection 3.3). In Section 4, we present the experimental sensing results obtained from our field measurements with different 5G signals. Finally, Section 6 provides the conclusion, summarizing our findings and outlining potential directions for future research.

2. OFDM-based Sensing Principle

The received signal $y(t)$, resulting from the transmission of $x(t)$ through a linear time-variant (LTV) radio channel [25,26], can be modeled as a sum of P reflected paths, indexed by $p = 1, \dots, P$, and a line-of-sight (LOS) path indexed by $p = 0$:

$$y(t) = \sum_{p=0}^P g_p x(t - \tau_p) e^{j2\pi f_p^{(D)} t}. \quad (1)$$

Each path is characterized by a complex attenuation coefficient g_p , a delay of τ_p seconds, and a Doppler frequency shift $f_p^{(D)}$ in Hz. For the LOS path and reflections from static objects, the Doppler shift $f_p = 0$. Equation 1 can be rewritten in integral form as:

$$y(t) = \int_{-\infty}^{\infty} h(t, \tau) x(t - \tau) d\tau, \quad (2)$$

which generalizes the Linear Time-Invariant (LTI) case, where $h(t, \tau)$ reduces to $h(\tau)$, and Equation 2 describes the received signal as the convolution of the CIR with the transmitted signal. The time-varying CIR can then be expressed as:

$$h(t, \tau) = \sum_{p=0}^P g_p e^{j2\pi f_p^{(D)} t} \delta(\tau - \tau_p). \quad (3)$$

Since paths resulting from reflections from moving objects exhibit non-zero Doppler shifts, the complex gain of each path oscillates as a function of time. These oscillations appear in the time-variant (TV) CIR at each delay tap $\tau = \tau_p$. To extract the Doppler shift frequency $f_p^{(D)}$ for the p -th path, the TV-CIR can be analyzed along the time dimension t using a Fourier transform. The resulting Doppler shifts can then be converted to velocities of the reflecting objects, while the time delays τ_p provide estimates of the total propagation path lengths.

In OFDM systems, the sampled received signal $y[n]$ is processed as follows. First, the stream is divided into segments corresponding to individual OFDM symbols (including cyclic prefix). After cyclic prefix removal, each symbol of length N (where N is the FFT size) is shaped into a two-dimensional matrix $y[n, s]$, where the first dimension represents time-

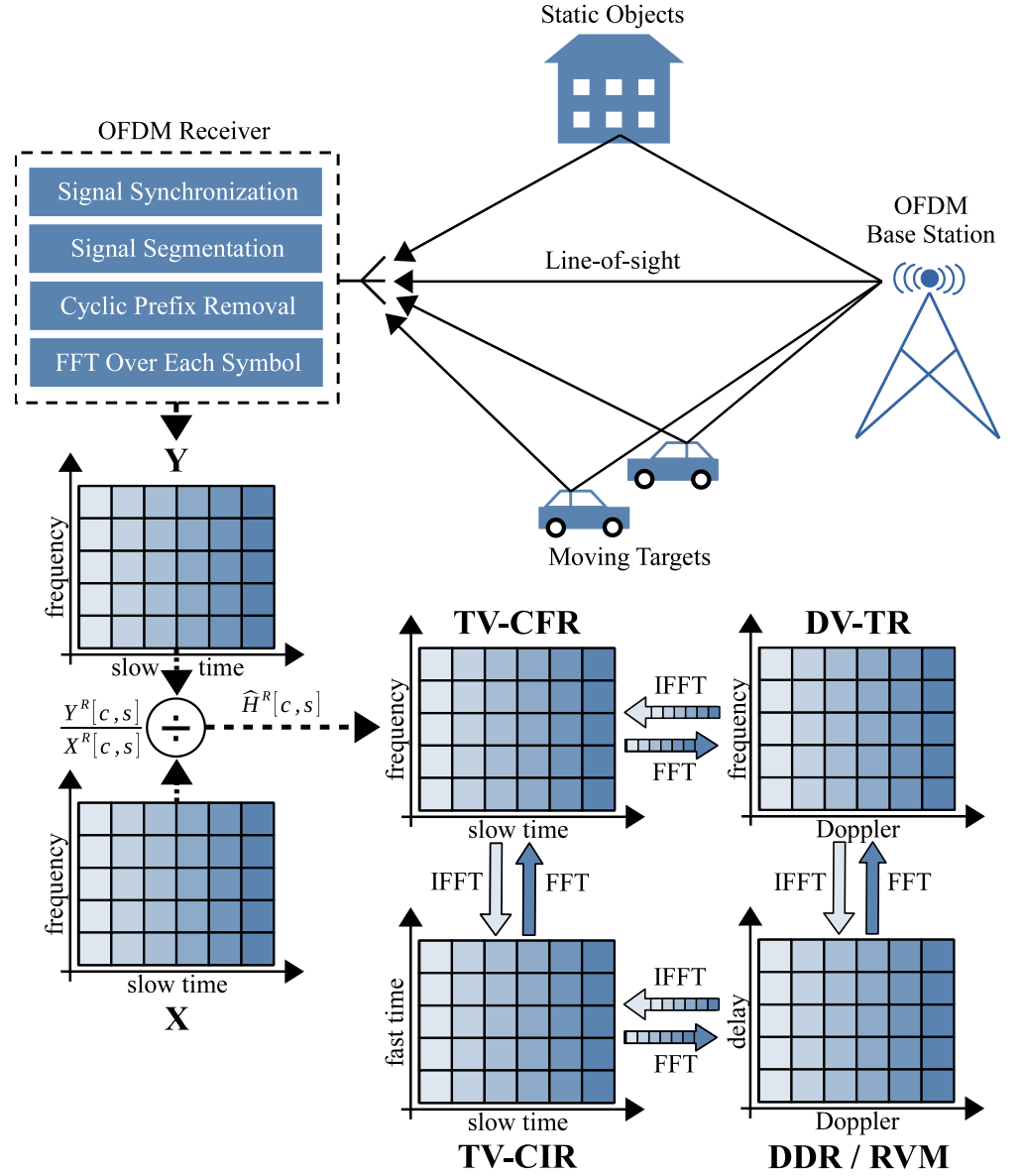


Figure 1. Description of the tested OFDM-based sensing scenario: matrix $\mathbf{Y}$ of received carrier states is decoded, based on real measurements, while matrix $\mathbf{X}$ of transmitted carrier states is assumed to be known (as in the base station).

domain samples within a symbol, and the second dimension represents the symbol index. Each symbol then undergoes a Fast Fourier Transform (FFT) to produce the OFDM resource grid $Y[c, s]$ in the frequency domain, where c denotes subcarrier index and s denotes OFDM symbol index. This process is explained in more detail using the 5G waveform example in Section 3.1. Channel estimation is essential for equalizing user data. For this purpose, known pilot sequences are embedded at predetermined positions R in the transmitted grid $X[c, s]$. The channel frequency response (CFR) at pilot locations is estimated using:

$$\hat{H}^{(R)}[c, s] = \frac{Y[c, s]}{X[c, s]}, \quad (c, s) \in R. \quad (4)$$

The full time-varying CFR estimate $\hat{H}[c, s]$ is then obtained by applying two-dimensional interpolation to $\hat{H}^{(R)}[c, s]$.

Since the TV-CFR is computed in every modern OFDM-based communication system, it becomes straightforward to integrate radar functionality using the same CFR samples, as

illustrated in bottom part of Fig. 1. First, the TV-CFR is obtained by dividing the received signal by the known transmitted (reference) signal values. Next, the TV-CIR is generated by applying an IFFT to each column of the CFR matrix. Finally, each row of the resulting CIR matrix is analyzed using an FFT to extract the frequencies of potential oscillations, yielding the delay–Doppler response (DDR). The axes of the DDR can be converted to range and velocity, forming a range–velocity map (RVM). Peaks in the RVM indicate the distance (in meters) and speed (in m/s) of moving objects near the receiver. It is also important to note that an equivalent result can be obtained by following an alternative processing path based on the Doppler-variant Transfer Res ponse (DV-TR).

3. 5G NR ISAC Receiver

3.1. Communication

Fig. 2 presents an example of the OFDM resource grid of a 5G waveform. The 5G NR signal is structured into 10 ms frames, each divided into 10 subframes of 1 ms duration [27, 4.3.1 Frames and subframes]. Each subframe is further divided into slots, determined by the numerology parameter μ [27, 4.3.2 Slots]. In this work, we set $\mu = 1$, corresponding to a subcarrier spacing (SCS) of 30 kHz and resulting in two 0.5 ms slots per subframe [27, 4.2 Numerologies]. Each slot consists of 14 OFDM symbols.

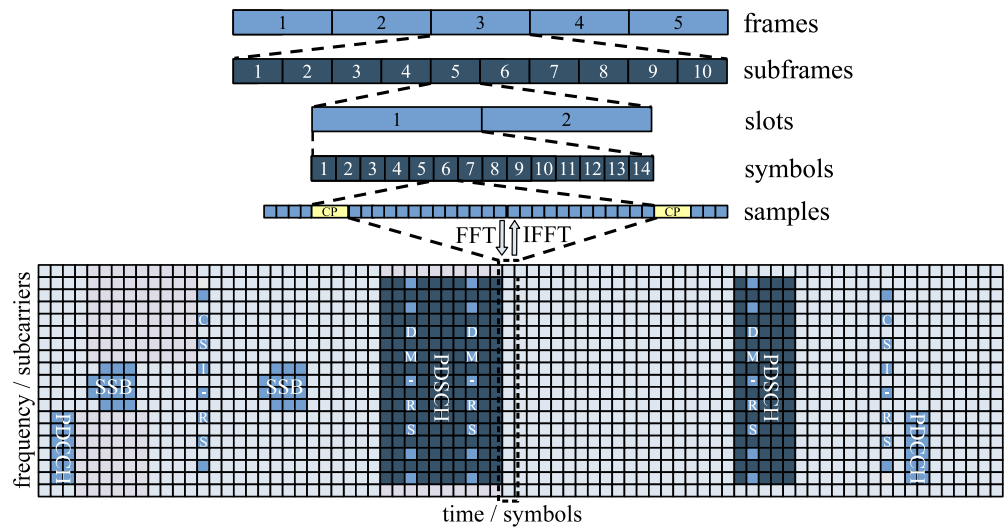


Figure 2. 5G NR resource time-frequency (OFDM symbols-carriers) grid for SCS=30kHz (description in the text)

The 5G OFDM resources can be represented on a two-dimensional grid, where the x-axis corresponds to time (consecutive OFDM symbols) and the y-axis corresponds to frequency (OFDM carriers). Each point on this grid is referred to as a resource element (RE) [27, 4.4.3 Resource elements] and is typically modulated using various orders of quadrature amplitude modulation (QAM) [27, 5.1 Modulation mapper] to convey information. Before transmission, each symbol (i.e., each column of the grid) undergoes an inverse Fast Fourier Transform (IFFT). Then, a portion of the end of the resulting time-domain vector is copied to its beginning to form a cyclic prefix (CP), which helps mitigate intersymbol interference (ISI). In 5G, the CP of the first symbol in certain slots is longer than that of the remaining symbols [27, 5.3 OFDM baseband signal generation]. Finally, all modulated symbols are concatenated to construct the complete time-domain signal.

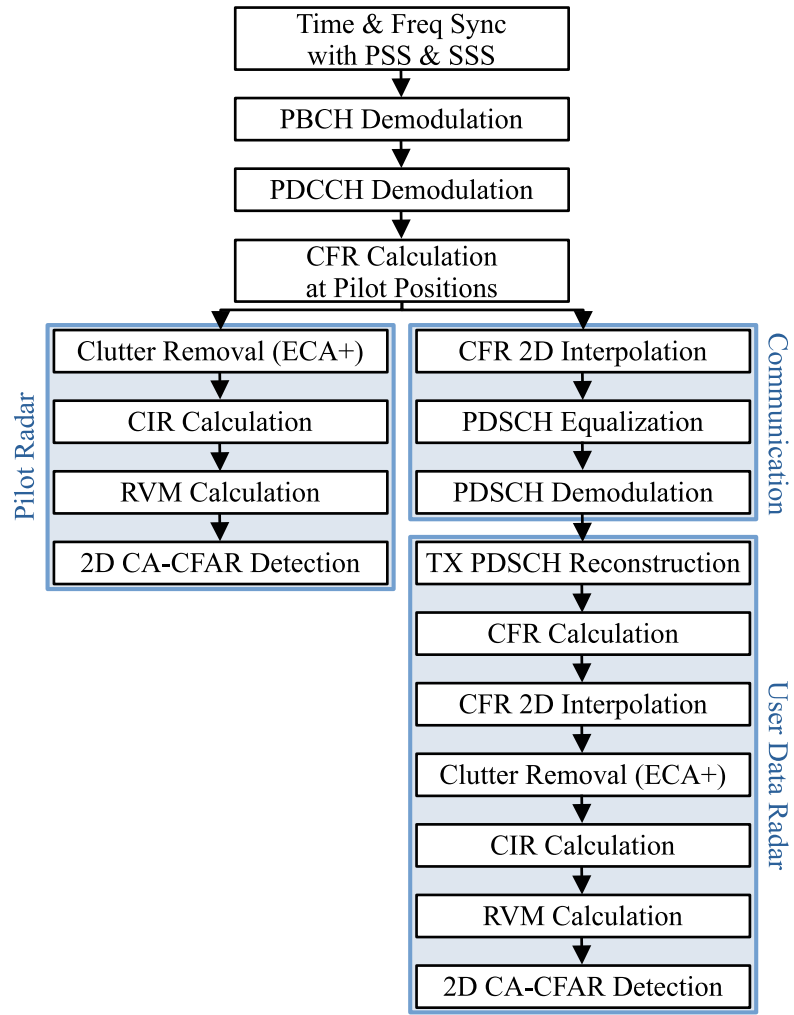


Figure 3. Operations performed in the 5G NR-based ISAC receiver

To successfully decode data, a 5G receiver must perform several steps to reverse the transmitter's modulation process. A simplified block diagram of these steps is shown in Fig. 3. The first task is time and frequency synchronization, which is accomplished using the Primary Synchronization Signal (PSS). The PSS is located in the first symbol (index 0) of the Synchronization Signal Block (SSB) [27, 7.4.3.1 Time-frequency structure of an SS/PBCH block] which is presented in Fig. 4. SSBs are built from 4 symbols occupying 240 subcarriers and are typically transmitted in bursts of size 1 to L_{max} . These bursts are periodically repeated, with a commonly used repetition period of 20 ms [28, 4.1 Cell search]. While their timing positions within this period are fixed, their frequency locations can vary depending on the Global Synchronization Channel Number (GSCN) raster [29, 5.4.3 Synchronization raster]. To achieve synchronization, the receiver correlates the known PSS sequences with the received signal to determine the SSB's frequency position, the coarse (integer) carrier frequency offset (CFO), and its time position within the received frame. The synchronization procedure is then refined by performing a more precise fine (fractional) CFO estimation using the CPs [30].

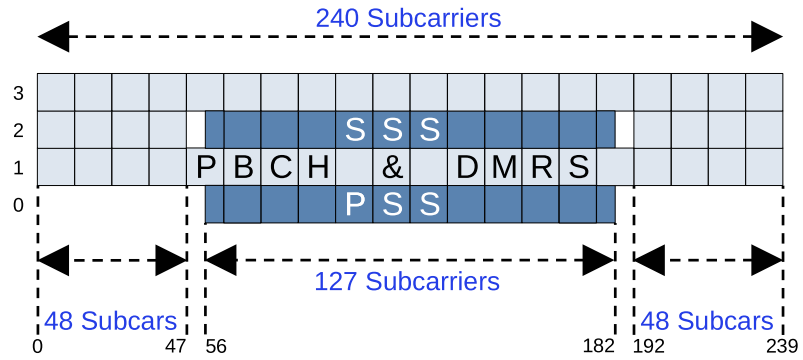


Figure 4. Structure of the synchronization signal block (SSB)

The next step is Physical Broadcast Channel (PBCH) demodulation. However, before this can be performed, the Cell ID must first be determined. This is accomplished using the Secondary Synchronization Signal (SSS), which is located in the third symbol (index 2) of the SSB [27, 7.4.3.1 Time-frequency structure of an SS/PBCH block]. The SSS transmits one of 336 binary m-sequences [27, 7.4.2.3 Secondary synchronization signal]. By performing a blind search, i.e., correlating the known SSS sequences with the received one, the receiver identifies the $N_{ID}^{(1)}$ value. Similarly, the $N_{ID}^{(2)}$ value can be determined using the PSS, which transmits one of 3 different m-sequences [27, 7.4.2.2 Primary synchronization signal]. The complete Cell ID is then computed as defined in [27, 7.4.2.1 Physical-layer cell identities]:

$$N_{CellID} = 3 \times N_{ID}^{(1)} + N_{ID}^{(2)} \quad (5)$$

The remaining resource elements (REs) of the SSB, are occupied by the PBCH and its associated Demodulation Reference Signals (PBCH DM-RS) [27, 7.4.3.1 Time-frequency structure of an SS/PBCH block]. After PBCH demodulation and Broadcast Channel (BCH) decoding [31, 7.1 Broadcast channel], the Master Information Block (MIB) can be extracted to obtain essential cell configuration information [32, 6.2.2 Message definitions]. Notably, the entire SSB is the only always-on signal in the 5G NR system.

Once synchronized and equipped with initial system information, the 5G receiver proceeds to acquire additional details regarding the exact locations of transmitted user data. This information is transported through the Physical Downlink Control Channel (PDCCH) [27, 7.3.2 Physical downlink control channel (PDCCH)]. Since the position of the PDCCH within the resource grid is not fixed, the receiver must perform blind detection across predefined PDCCH Search Spaces, which specify potential regions where the PDCCH may be located [28, 10.1 UE procedure for determining physical downlink control channel assignment]. The PDCCH carries Downlink Control Information (DCI), which contains instructions about the physical resource allocation of user data [31, 7.3 Downlink control information]. To successfully decode the DCI, the receiver must know the Radio Network Temporary Identifier (RNTI), as the Cyclic Redundancy Check (CRC) of each DCI message is scrambled using this number [31, 7.3.2 CRC attachment]. There are several types of RNTIs [33, 7.1 RNTI values]: some are common RNTIs, known to all users and used to scramble DCIs containing broadcast or system-wide information; others are user-specific RNTIs, which are known only to individual users, allowing them to decode their respective DCIs and identify the locations of their assigned data [33, 5.1 Random Access procedure]. Finally, once the DCI is successfully decoded, the receiver obtains the necessary information to access the user data, which is transmitted through the Physical Downlink Shared Channel (PDSCH) [27, 7.3.1 Physical downlink shared channel].

Unlike LTE, where Cell-Specific Reference Signals (CRS) were continuously transmitted at fixed positions across the entire OFDM grid, 5G does not include any always-on

reference signals for channel estimation. Instead, each physical channel defines its own set of Demodulation Reference Signals (DM-RS), which are transmitted among other symbols, for example, PBCH DM-RS, PDCCH DM-RS, and PDSCH DM-RS [27, 7.4.1 Reference signals]. Before demodulating the channel symbols, the receiver estimates the CFR using these pilots. Then it is interpolated over the remaining REs and used for equalization, i.e., to mitigate the channel's impact on the received data. Another type of reference signal present in the 5G waveform is the Channel State Information Reference Signal (CSI-RS) [27, 7.4.1.5 CSI reference signals]. It is transmitted by the gNodeB to enable estimation of downlink radio channel quality. Again, the key difference from the LTE CRS is that the 5G CSI-RS is configured per device rather than per cell, meaning it is not continuously transmitted and may not always be present in the signal.

3.2. OFDM-based Sensing

The Channel Frequency Response (CFR) estimates, which are already computed in every 5G communication device, can be repurposed for sensing applications. This concept is illustrated in Fig. 3, where, after the initial three processing steps, the diagram branches into two directions: one for standard communication processing and the other for radar. In the sensing branch, the first stage is clutter removal. Clutter appears as strong peaks near the zero-Doppler region, originating from the LOS path and reflections from static objects such as buildings or trees. Since clutter often has significantly higher energy than weaker target reflections, it can mask moving targets and must therefore be removed from the RVM. In our work, we tried several variants of the Extensive Cancellation Algorithm (ECA) [34], specifically ECA by Carrier and Doppler (ECA-CD) [35] and ECA+ [36,37]. Ultimately, we selected ECA+ due to its low computational complexity and strong clutter suppression performance. Once clutter is removed, additional OFDM-based radar processing, as described in Section 2, is applied. In our previous studies [16,17], we demonstrated that this approach, using CFRs derived from 5G CSI-RS and our custom RoD pilots, can effectively detect moving vehicles. Since the CFR in this method is computed using pilot signals, we refer to it as Pilot Radar, and the resulting RVM as the Pilot RVM (P-RVM).

During signal reception, the 5G receiver demodulates and decodes various parts of the waveform, as described in Section 3.1. This enables reconstruction of the transmitted symbols at those locations, allowing them to be reused as pilot-like symbols for computing a denser CFR for sensing purposes. This process is illustrated in the bottom-right section of Fig. 3. However, unlike the pilot-only CFR, this reconstructed CFR is non-uniformly sampled, since the positions of all physical channels and signals, particularly the PDSCH, may appear random from a radar perspective. To address this, two-dimensional interpolation for non-uniformly distributed samples is applied to fill in the missing values. In this work, a linear interpolation method is used [38]. After interpolation, the standard OFDM-based sensing procedure is performed, including clutter removal and RVM generation. The resulting RVM, referred to as the User Data RVM (UD-RVM), is constructed from a significantly larger number of samples than the P-RVM, resulting in improved sensing performance. In our previous work [24], we presented simulation results demonstrating that the use of UD-RVMs enhances detection accuracy and enables the detection of high-speed targets that cannot be captured using P-RVMs.

3.3. CAF-based Sensing

CAF [8]

[?]

cross ambiguity function (CAF), that is, a widely used CAF-based passive radar technique [8].

$$\text{CAF}(R_b, V_b) = \int_0^{T_i} y(t) \hat{x}^* \left(t - \frac{R_b}{c} \right) \exp \left(-j \frac{2\pi}{\lambda} V_b t \right) dt, \quad (6)$$

where $y(t)$ is a continuous-time version of the received signal (??), $\hat{x}(t)$ is an estimate of the transmitted signal: $y(t)$ demodulated, decoded and back-modulated or a signal from the second/reference antenna looking towards a transmitter, z^* denotes the complex conjugation of z , T_i is the integration time, R_b is the bistatic range, V_b is the bistatic velocity, and $\lambda = c/f_c$ is the wavelength (c - speed of light, f_c - carrier frequency of the 5G/6G service).

4. Field Measurements

4.1. 5G NR Configuration and Antenna Geometry

Our field measurements intended to experimentally validate the proposed OFDM-based approaches that utilize different combinations of control, reference and user data, and to compare their performance with the conventional CAF-based radar. During the experiments, we transmitted three different 5G signals generated using the MATLAB 5G Toolbox. The common components of these signals included two SSBs per SS Burst, transmitted every 20 ms, and a PDCCH transmitted every five frames. The CSI-RS and PDSCH configurations of these signals are summarized in Table 1, and they are also illustrated in Fig. 5.

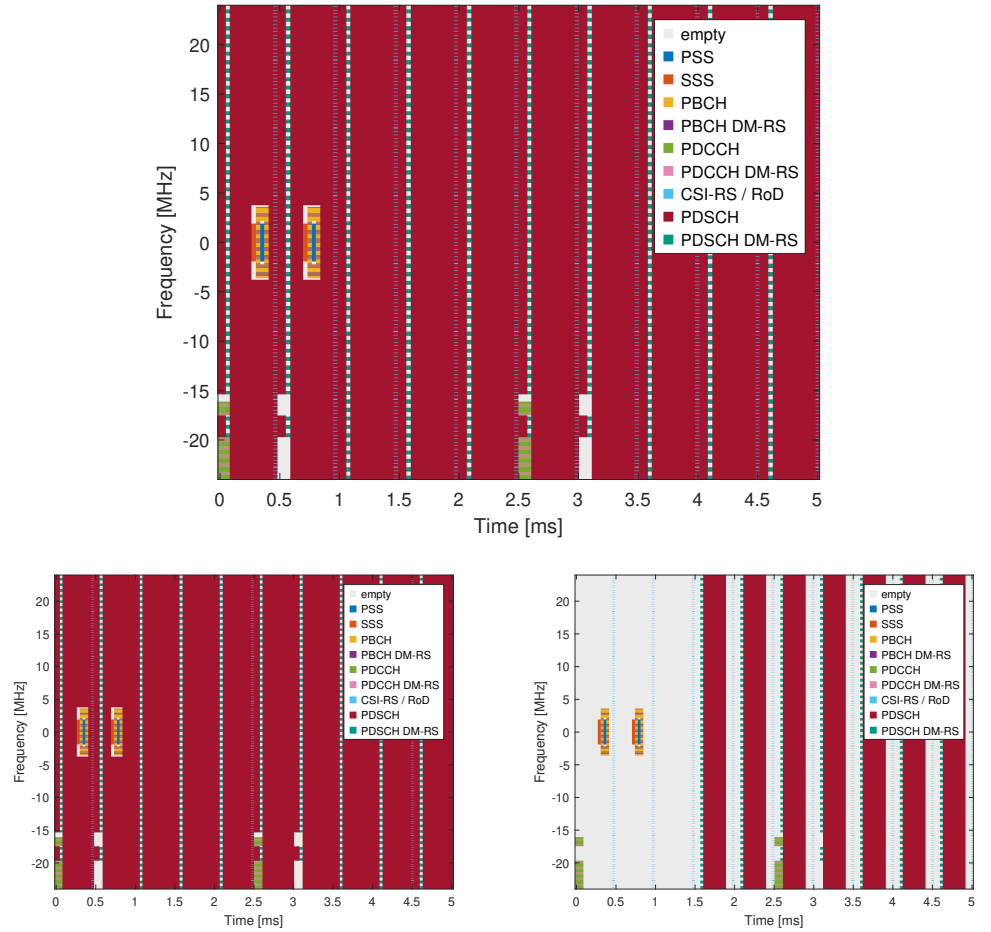
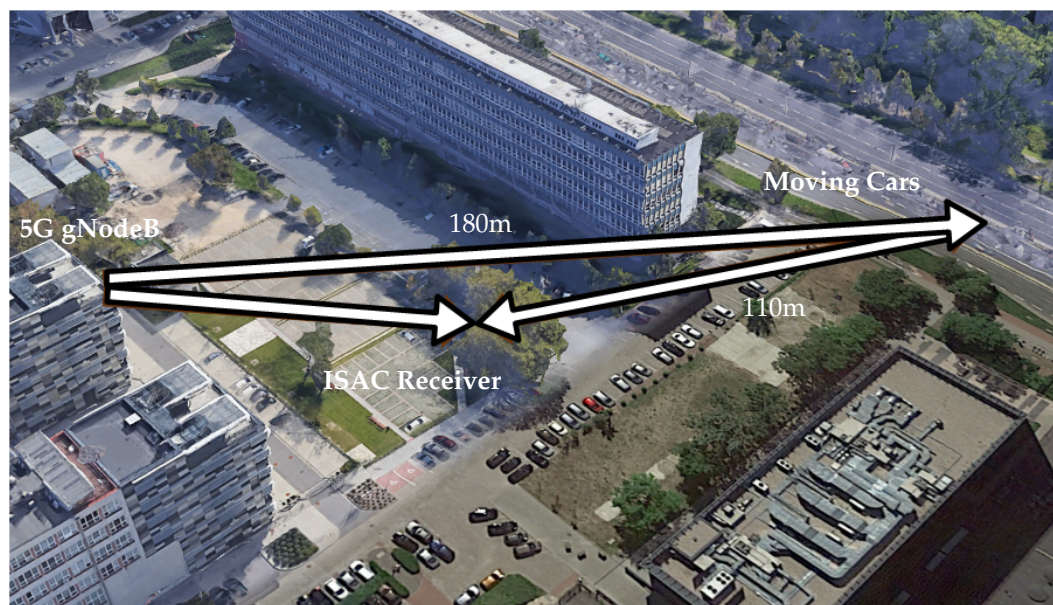


Figure 5. 5G signals used in field measurements.

Table 1. CSI-RS time and frequency densities used in test signals A, B and C

Signal	CSI-RS Time Density	CSI-RS Freq Density	PDSCH Allocation
A	14	12	All remaining Resource Elements
B	56	12	All remaining Resource Elements
C	14	12	Slots [3 - 40], Symbols [3 - 13]

The field measurement setup was identical to that used in our previous work [17] and is illustrated in Fig. 6. The base station was emulated using a software-defined radio (SDR) positioned approximately 20 m above the ground. Before transmission, the signal was amplified using a high-power amplifier and transmitted through an antenna with a beamwidth of approximately 16° in both the horizontal and vertical planes and a directivity of about 19 dBi. We operated at a carrier frequency of 5.8 GHz, which lies within an unlicensed band, and used a signal bandwidth of approximately 26 MHz. All relevant 5G NR parameters used during the experiment are summarized in Table 2.

**Figure 6.** Field measurements geometry.**Table 2.** 5G NR experiment parameters (P-ilot (synchro), C-ontrol, D-ata)

Parameter	Value	Comment
N_{fft}	2048	FFT length
$N_{\text{cp1}}, N_{\text{cp2}}$	144, 176	Length of cyclic prefixes
Modulation	BPSK (P), 4-QAM (P, C, D)	Type of carrier modulation
f_c	5.8 [GHz]	Carrier frequency
B	26 [MHz]	Bandwidth of the signal

The ISAC receiver, placed in the parking lot near the high-speed road, acquired the signal after it passed through a chain of low-noise amplifiers and bandpass filters between the antenna and the SDR, with a total amplifier gain of 20 dB. The SDR parameters are summarized in Table 3. The distance between vehicles and receiver was approximately 110 m, while the distance between the vehicles and the transmitter was about 180 m. Therefore, the total bistatic range [8] is estimated to be around 240 m.

Table 3. Parameters of the SDR and PC platform

Parameter	Specification
SDR Type	NI USRP 2953R (USRP X310)
SDR Daughterboards	UBX-160
SDR Synchronization	GPS Disciplined Oscillator
SDR Interface	PCI-Express x4

4.2. Radar Configurations

Each signal captured by a receiver was segmented into 100 ms (10-frame) intervals with a 50 ms (5-frame) step. Each segment was first synchronized using the PSS, followed by cell identification using a combination of the PSS and SSS. Subsequently, the PBCH was demodulated, and the MIB was decoded. To ensure correct signal reception, the BCH CRC was also verified. Since the main focus of this study is on radar performance, the subsequent communication stages were simplified. Instead of performing a full PDCCH Search Space decoding and using DCI to locate PDSCH, we utilized a priori knowledge of PDCCH and PDSCH positions. However, unlike many simulation-based studies, we did not assume that PDSCH symbols were known to the receiver, as would be the case in a real 5G network. Instead, we demodulated the received QAM symbols, reconstructed the transmitted signal, and calculated the BER by comparing the reconstructed symbols with the originally generated ones.

This approach inherently assumes knowledge of all RNTIs present in the network, since such information is required for the receiver to identify all PDSCH locations. This assumption holds only in a single-user scenario, where the ISAC receiver is the only user in the network, an idealized and unrealistic case. In practical multi-user networks, user-specific RNTIs are unknown, making it impossible to locate other users PDSCHs during runtime (although an offline blind search of all possible RNTIs might be possible). Therefore, in this study, we compare the sensing performance of OFDM-based radars that exploit different combinations of channels and signals available in the 5G NR downlink:

1. **SSB Radar (S-Radar, S-RVM):** Uses only SSBs (PSS, SSS, PBCH, and PBCH DM-RS), the only always-on downlink signals, to compute the CFR (2D linear interpolation).
2. **SSB & Control Radar (SC-Radar, SC-RVM):** Uses SSB together with PDCCH and its DM-RS, which are typically present in most 5G NR downlink configurations, to compute the CFR (2D linear interpolation).
3. **Pilot Radar (P-Radar, P-RVM):** Uses only CSI-RS signals, which are an optional part of the 5G NR downlink waveform, to compute the CFR (no interpolation required).
4. **SSB, Control & Pilot Radar (SCP-Radar, SCP-RVM):** Uses SSB, PDCCH with PDCCH DM-RS, and CSI-RS to compute the CFR (2D linear interpolation).
5. **SSB, Control, Pilot & Data Radar (SCPD-Radar, SCPD-RVM):** Uses all downlink channels and signals (SSB, PDCCH with PDCCH DM-RS, CSI-RS and PDSCH with PDSCH DM-RS to compute the CFR (2D linear interpolation).
6. **SSB, Control & Data Radar (SCD-Radar, SCD-RVM):** Uses all downlink channels and signals except CSI-RS (i.e., SSB, PDCCH with PDCCH DM-RS, and PDSCH with PDSCH DM-RS) to compute the CFR (2D linear interpolation).

For clarity, these configurations are also summarized in Table 4. We also analyze the SCD-Radar performance when only a subset of PDSCH locations is known to the receiver, representing a more realistic multi-user scenario, in section 4.6.

Fig. 7a–f shows the RVMs obtained from radar processing of Signal A at a timestamp of 0.7 s for each configuration. The first two plots (a and b) demonstrate that both S-Radar and SC-Radar provide too few CFR samples to reliably detect targets. This issue persists

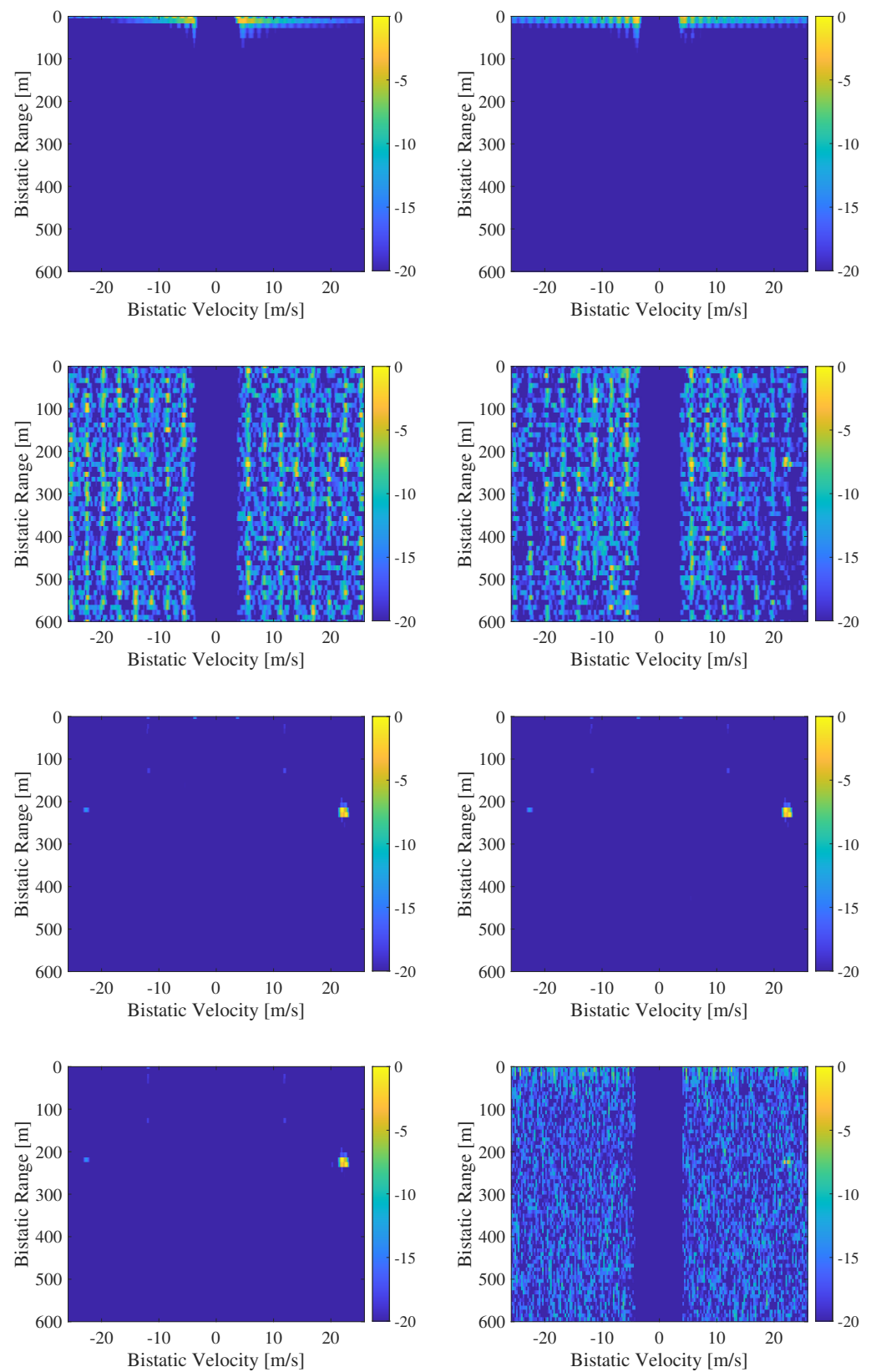


Figure 7. RVMs after processing of signal A at timestamp 0.7s with a) S-Radar, b) SC-Radar, c) P-Radar, d) SCP-Radar, e) SCPD-Radar, f) SCD-Radar, g) GT-Radar, h) CAF-Radar *na rys. słabo widać też dwa obiekty koło 100m, prędkość -12m/s, 12 m/s; może wziąć je w żółte/czerwone kółka i coś napisać w tekście?*

Table 4. Specification of tested OFDM-symbol based radar configurations

Radar Denotation	SSB	PDCCH & PDCCH DM-RS	PDSCH & PDSCH DM-RS	CSI-RS	2D Interpolation
S-Radar	Yes	No	No	No	Yes
SC-Radar	Yes	Yes	No	No	Yes
P-Radar	No	No	No	Yes	No
SCP-Radar	Yes	Yes	No	Yes	Yes
SCPD-Radar	Yes	Yes	Yes	Yes	Yes
SCD-Radar	Yes	Yes	Yes	No	Yes

across other timestamps; therefore, additional results for these two configurations are omitted. When using only CSI-RS (plot c) or CSI-RS combined with SSB and PDCCH (plot d), a single target peak becomes visible in the RVM; however, these maps remain noticeably noisy. This situation improves significantly for the SCPD (plot e) and SCD (plot f) radars, where the addition of user data greatly increases the number of CFR samples. For these configurations, not only is the noise level reduced, but a second weaker target also becomes detectable in the RVM.

4.3. Ground Truth Values

In the measurements, our objective was to detect vehicles moving along the high-speed road. Since the targets were non-cooperative, their exact positions and velocities were unknown. To establish a performance baseline, we applied additional processing at each timestamp using an OFDM-based radar configuration in which all RE positions and values are known (i.e., assuming perfect synchronization and a priori knowledge of the transmitted signal). Although such a scenario is unrealistic in practical bistatic systems, it provides an effective ground truth for evaluating other methods. We therefore refer to this configuration as the Ground Truth Radar (GT-Radar). For comparison, we also performed classical CAF processing. This method is well-established and had been previously validated in [reference] at the same location and under similar conditions, which confirms its reliability. The resulting RVMs for both GT-Radar and CAF-Radar are shown in Fig. 7g-h.

After computing the RVMs, we applied a two-dimensional Cell-Averaging Constant False Alarm Rate (CA-CFAR) detector [39] to identify potential targets. The algorithm produces a binary detection map for each RVM sample, indicating target presence or absence, and also estimates the local noise floor around each detected peak. For each timestamp, we first selected the GT-Radar CFAR detection corresponding to the highest peak energy. Since our goal was to detect vehicles within a specific region of interest, we filtered the CFAR detections using the following constraints:

1. Bistatic velocity within either $[-30\text{m/s}, -10\text{m/s}]$ or $[10\text{m/s}, 30\text{m/s}]$
2. Bistatic range within $[140\text{m}, 280\text{m}]$

The resulting filtered detections for each signal and each timestamp are shown in Fig. 8.

To further minimize false alarms, we introduced an additional consistency check. A detection is considered valid only if the same object is observed in at least two consecutive timestamps. Two detections are treated as originating from the same object when both of the following conditions are met:

1. The difference in bistatic velocity is below 2 m/s
2. The difference in bistatic range does not exceed 30 m

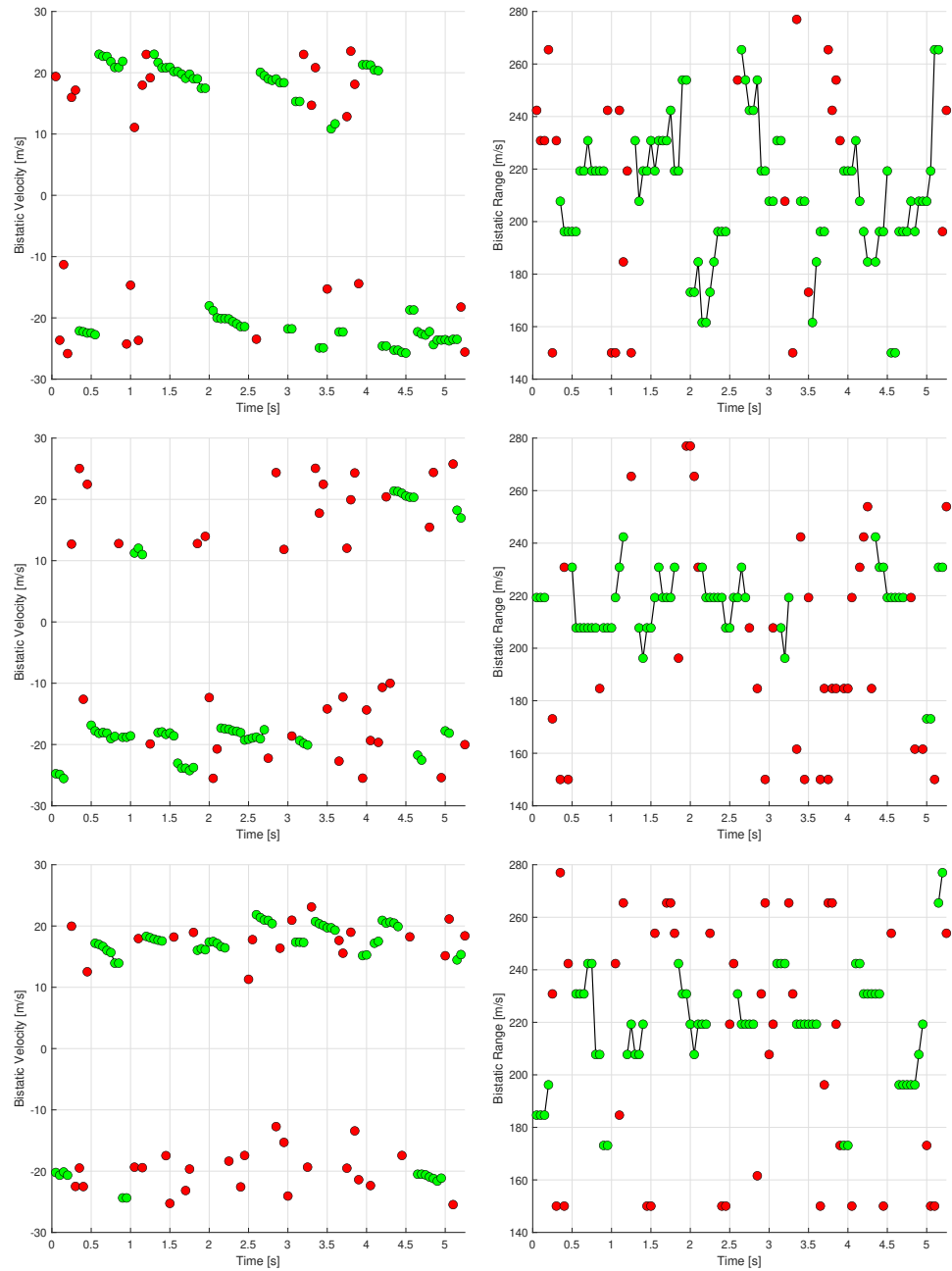


Figure 8. Bistatic velocity and distance of detected targets using GT-Radar. Green - assumed correct detection. Red - assumed false detection.

In Fig. 8, validated detections are shown in green, while spurious ones are marked in red. Points associated with the same moving object are linked by straight lines. As illustrated, correct detections form smooth, continuous trajectories in both velocity and range, whereas false alarms are scattered irregularly. This behavior confirms that the applied consistency rule reliably isolates true targets. In the remainder of this section, only timestamps in which the GT-Radar produces valid detections (green points) are used for evaluating the performance of the remaining methods.

4.4. Radar Performance Comparison

To evaluate the performance of each sensing method, we used the Peak-to-Noise Floor Ratio (PNFR), a metric representing the relative height of the detected peak. Given the CFAR-estimated noise level A_{noise} and the amplitude of the detected peak A_{echo} , PNFR is defined as:

$$PNFR = 20 \log_{10} \left(\frac{A_{echo}}{A_{noise}} \right) \quad (7)$$

For every sensing configuration, the CFAR detector, with the probability of false alarm (PFA) set to 10^{-3} , was applied to the corresponding RVM at all timestamps where a ground-truth detection was available. The resulting peak's bistatic velocity and range were then compared against the ground-truth values. If the detection matched the true target, its PNFR was calculated. Fig. 9 summarizes the PNFR values obtained through this procedure. The results reveal a clear trend: a larger number of CFR samples generally leads to higher PNFR.

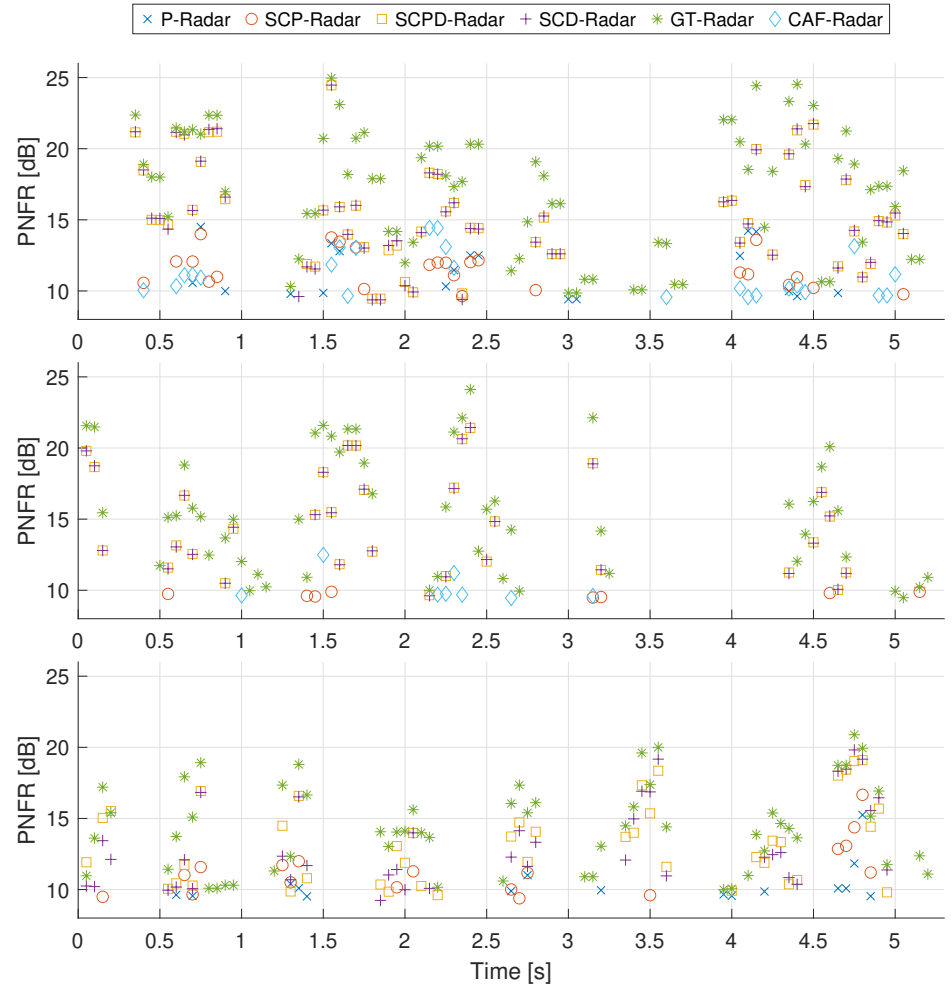


Figure 9. Signals A-C (from top to bottom)

For Signals A and B, where the PDSCH occupies most of the available REs, the difference between the SCPD- and SCD-Radars is negligible. In these cases, adding a small number of CSI-RS samples does not significantly influence performance. In contrast, for Signal C, where some slots contain no PDSCH, the additional CSI-RS pilots help to stabilize the interpolation in those regions, resulting in noticeable PNFR improvement. This behavior is examined in greater detail in Section 4.6. We also observe that the PNFR values of SCD- and SCPD-Radars remain slightly below those of the Ground Truth Radar, even though they rely on a similar number of CFR samples. This degradation results from transmission errors in the demodulated user data, and is further analyzed in Section 4.5. Finally, the P- and SCP-Radars exhibit substantially lower PNFR, as expected due to the limited number of CFR samples they use.

To better highlight the differences between the radar configurations, we also evaluated the PNFR deviation from the ground truth for each method. Fig. 10 present the average PNFR differences for Signals A, B, C and aggregated across all datasets. These plots demonstrate that SCPD- and SCD-Radars deliver the strongest performance, with PNFR values within 2 dB of the ground truth. In contrast, radars relying primarily on pilot signals (P-Radar and SCP-Radar) show substantially lower performance, with PNFR deficits of approximately -3 to -6 dB and -5 to -7 dB, respectively. Nevertheless, adding even a small amount of additional information, specifically SSB and PDCCH, provides a noticeable improvement of 1-2 dB, confirming the benefit of utilizing more CFR samples for radar.

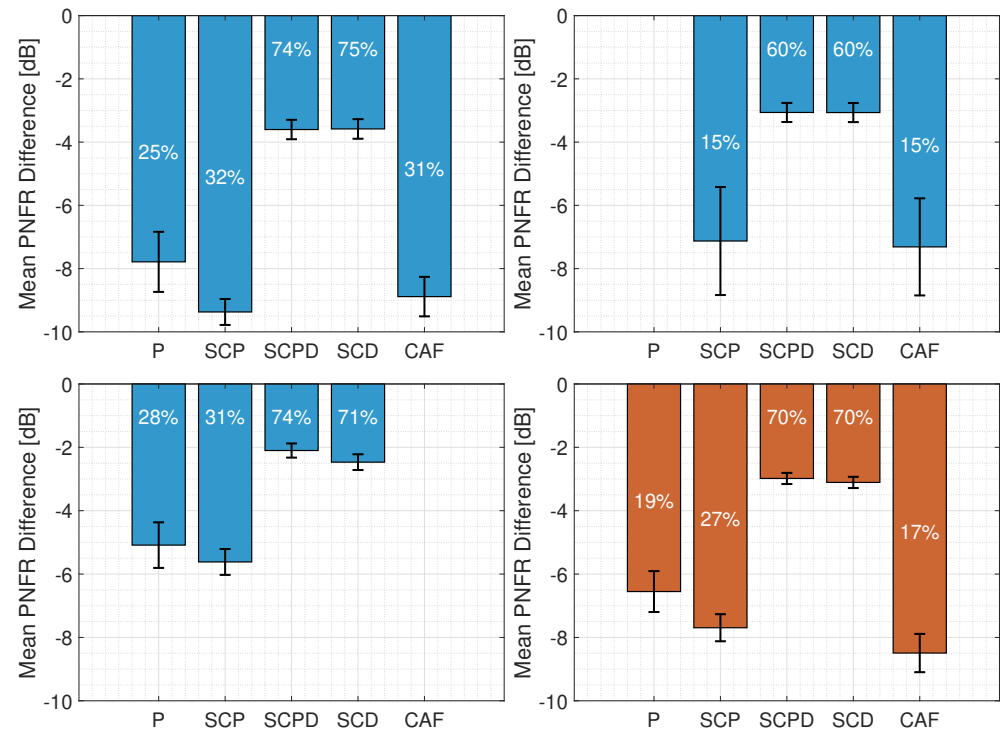


Figure 10. Radar Types vs Ground Truth

The same trends appear in the probability of detection (POD), defined as the number of correctly detected target divided by the total number of detections made by the GT-Radar. Table 5 shows how many timestamps contained a correct target detection for each radar type. The corresponding POD values are displayed on each bar in Fig. 10. SCPD- and SCD-Radars correctly detected the same targets in Signals A and B. In Signal C, they also detected targets in the same 27 timestamps. Beyond that, the SCPD-Radar achieved 6 additional correct detections (timestamps 0.55, 2, 2.1, 2.65, 3.6, 4.35), while the SCP-Radar achieved 2 additional correct detections (timestamps 3.4, 5.15). This indicates that using

additional CSI-RS samples provides a slight improvement when the signal contains fewer user-data resources. It is also important to note that in Signal B, where CSI-RS is very sparse (present only every 56 OFDM symbols), the P- and SCP-Radars were unable to detect high-speed targets, resulting in the absence of plot bars for these radars in the figure.

Table 5. Number of correct object (vehicle) detection for different radar types

Signal	P	SCP	SCPD	SCD	CAF	GT
A	16	20	49	49	14	66
B	0	0	29	29	2	41
C	5	11	33	29	0	47
All	21	31	111	107	16	154

4.5. BER Impact on Radar Performance

In a bistatic setup, the transmitted user data contained in the PDSCH is not known at the receiver. Therefore, we demodulated the PDSCH symbols, compared the reconstructed bits with the originally generated bitstream, and computed the BER as:

$$\text{BER} = \frac{N_{\text{errors}}}{N_{\text{bits}}}. \quad (8)$$

Fig. 11 presents the BER obtained at each timestamp of each signal. The values range from 10^{-2} to 10^{-1} , with every signal containing segments of both low (good) and high (poor) BER. These variations arise from changing propagation conditions during the measurements. Since the signals were transmitted sequentially, each one experienced a slightly different channel realization, leading to distinct BER fluctuations.

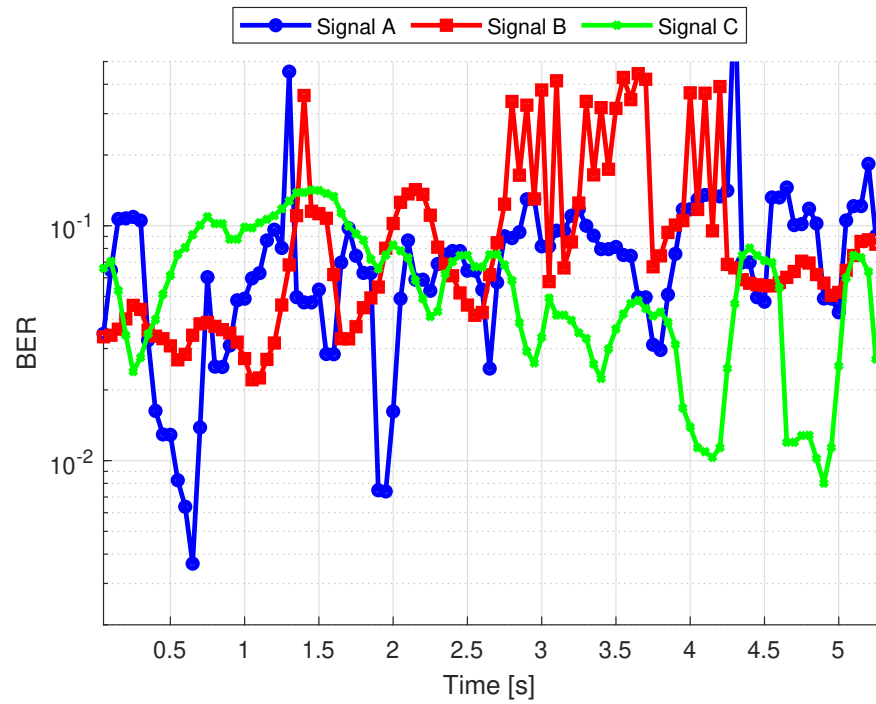


Figure 11. BER of user data bits (PDSCH) transmission with signals A, B and C.

Since the sensing stage, in SCPD- and SCD-Radars, treats the demodulated user bits as pseudo-pilots, without any knowledge of the actual BER, a high error rate may decrease radar performance. To evaluate this effect, we used the aggregated PNFR difference, relative

to the GT-Radar, across all signals and timestamps, obtained with the SCD- and SCPD-Radar. GT-Radar, represents the ideal case with BER = 0, as it assumes perfect knowledge of the transmitted bits. Next, the BER range was segmented into several intervals. Each PNFR difference value was assigned to the corresponding BER interval based on the measured BER at that timestamp. For every interval, we computed the average PNFR difference. The resulting relationship between BER and PNFR degradation is presented in Fig. 12.

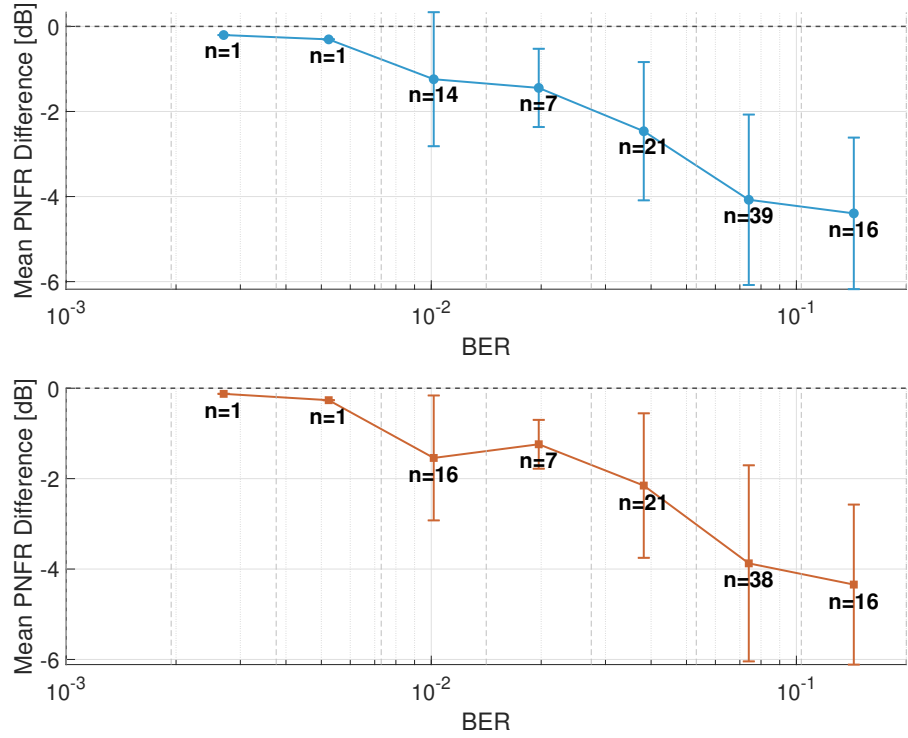


Figure 12. BER vs PNFR difference between GT-Radar and SCD (top) ; (SCPD) (bottom)

These results show that for low BER values, both SCD- and SCPD-Radars achieve performance only slightly worse than the GT-Radar, with differences below 0.5 dB. When the BER increases to around 2%, the curves begin to decline, and a significant performance degradation (approximately 3 dB) appears once the error rate reaches about 15%. It is also visible that the addition of CSI-RS signals in the SCPD-Radar provides a slight performance advantage in high-BER conditions. It is also important to note that the obtained values are relatively high compared to those typically observed in real 5G deployments. This is primarily due to two factors. First, our field measurements were performed using SDR hardware, which has significantly lower RF performance than commercial 5G base stations. Second, our communication processing chain is simplified; for instance, we omit Forward Error Correction (FEC), which plays a crucial role in maintaining low BER in practical 5G systems.

4.6. PDSCH fragments

Since having full knowledge of all PDSCH allocations and their DM-RS is unrealistic in a real 5G deployment, we investigated scenarios where only a portion of the PDSCH positions is known to the ISAC receiver, similar to the case where the receiver is connected to the network and decodes only assigned or shared resources. To emulate this, the received 5G OFDM grid ($1596 \times ??$ REs) was partitioned into ?? equally sized fragments of size ($?? \times ??$). A subset of these fragments was then randomly selected, and only the PDSCH

resources within these fragments were demodulated, reconstructed, and used, together with their DM-RS, for radar sensing. Figure 13 shows the results of this experiment for signal A at timestamp 0.7 s, where BER is relatively low (about 2%), and at timestamp 4.1 s, where BER is high (about 15%). For both SCPD- and SCD-Radars, we randomly selected a given percentage of fragments (10%, ..., 90%) 50 times, performed sensing, and compared the outputs with the ground-truth values. For each bar, the POD is printed in white for the SCPD-Radar and in black for the SCD-Radar, while the bar height corresponds to the mean PNFR of runs with successful detection.

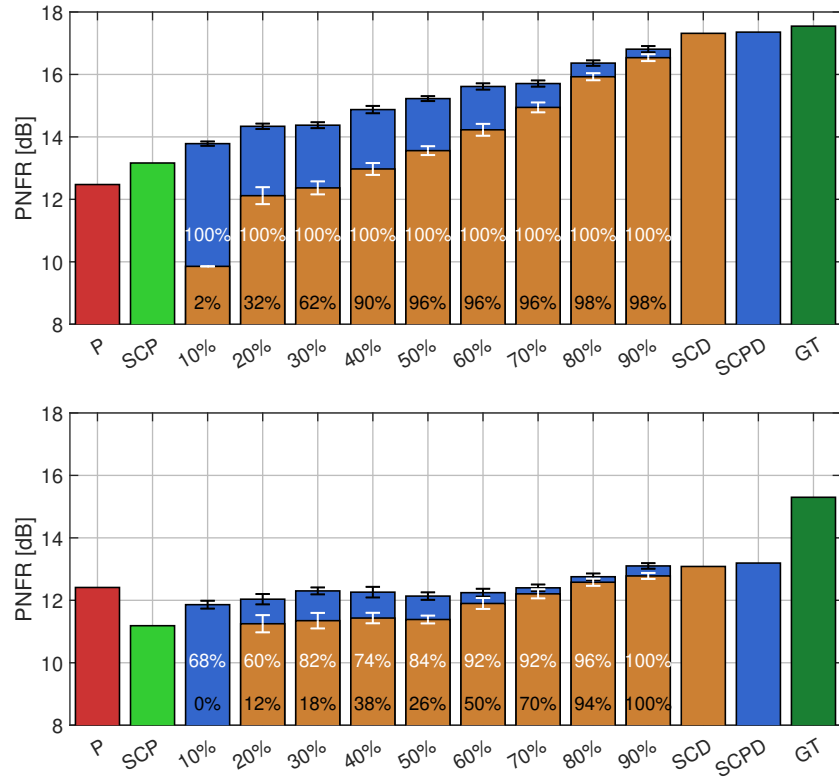


Figure 13. Signal A timestamp 0.7 i 4.1

The results for the 0.7 s timestamp show that increasing the amount of user data available to the radar consistently improves PNFR. The SCPD-Radar also achieves perfect POD, detecting the target in all iterations. Without CSI-RS, the SCD-Radar has too few samples to reliably detect the target when the number of known PDSCH positions is low (10–30%). However, once 40% or more of the positions are known, the POD exceeds 90%, which is a good result. The 4.1 s timestamp represents one of the few cases where the P-Radar, despite having fewer samples, performs better than the SCP-Radar. This is likely due to inaccurate interpolation: in such conditions, adding additional but erroneously (BER = 15%) reconstructed user symbols does not significantly improve sensing performance and may even slightly degrade it when the number of known PDSCH symbols is small.

5. Radar Sensitivity to User Data QAM Order

All signals transmitted during our field measurements used 4-QAM modulation for PDSCH. However, real 5G deployments use adaptive QAM orders depending on the Modulation and Coding Scheme (MCS), which is adjusted dynamically based on channel quality to keep BER low. Higher-order QAM constellation states have different amplitude, making them more sensitive to noise and potentially affecting radar performance. To complement our field results, we therefore performed simulations to assess how increased QAM order

influences sensing accuracy. To keep the simulation environment realistic, we exported the measurement location from OpenStreetMap and imported it into MATLAB. The transmitter, receiver, and carrier parameters were configured to match the field setup. Next, raytracing was performed using MATLAB's built-in functions [citation], producing propagation paths between transmitter and receiver along with their characteristics (PathLoss, PhaseShift, PropagationDelay). The simulation geometry and the extracted paths are shown in Fig. 14.

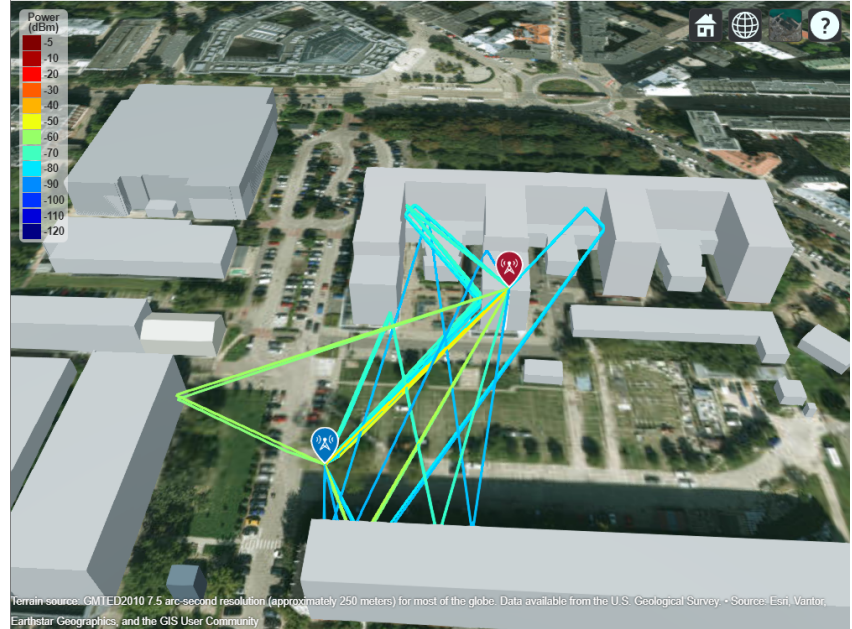


Figure 14. Simulation geometry.

Because the exported geometry does not include moving vehicles, we manually inserted additional dynamic path representing moving object. Its parameters, delay, Doppler, and complex gain, were taken from the strongest target peak observed in the RVM of Signal A at timestamp 0.7 s (Fig. 7). Using the complete set of paths, we generated the received signal according to eq. 1. We simulated Signal A using both 4-QAM and 256-QAM, and added AWGN matching the noise level measured in the field. Finally, radar processing was performed using GT-Radar. Fig. 15 presents the resulting radar outputs.

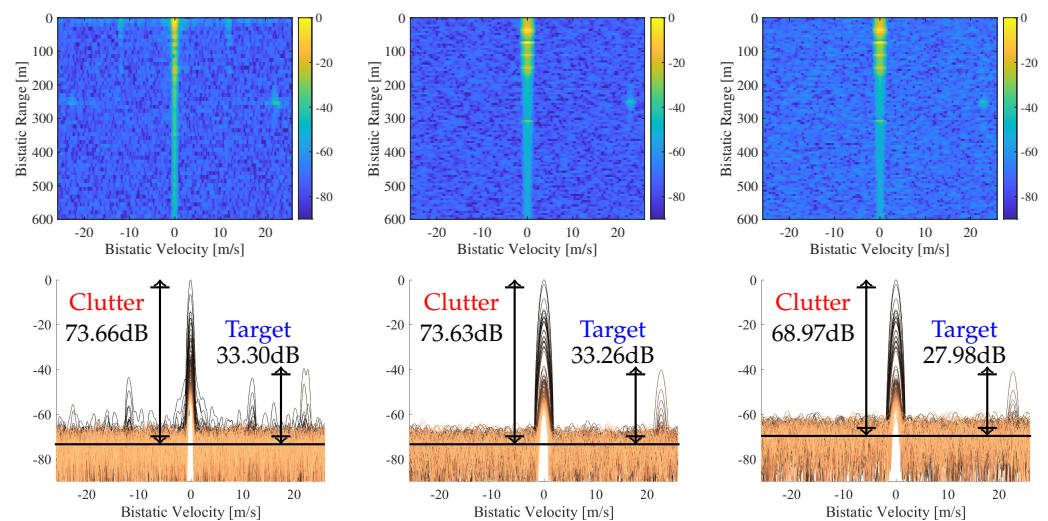


Figure 15. RVMs and Doppler profiles after processing of signal A: field measurements with 4-QAM (left); simulations with 4-QAM (middle); simulations with 256-QAM (right).

The results indicate that the PNFR difference between using 4-QAM and 256-QAM is approximately 4.5 dB, which is a relatively small degradation. It is important to note that these results were obtained using GT-Radar, where all transmitted user bits are known a priori. Under the high-noise conditions in this simulation, SCPD- and SCD-Radars cannot operate because the transmission BER is extremely high (about 38%), making symbol reconstruction impossible. However, real 5G systems are designed to avoid such situations: the MCS is continuously adapted to maintain low BER, meaning that 256-QAM would never be selected under these poor channel conditions. In summary, when channel quality degrades, the 5G network will reduce the QAM order, which in turn improves radar performance. Conversely, when the channel quality is good, higher QAM orders are used, leading to slightly worse radar performance compared to the 4-QAM case. However, in those conditions the noise floor is also lower, enabling the radar to operate reliably despite the higher QAM order.

6. Conclusions

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